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Reference

- video

OpenCourseWare - Mathematics for Computer Science - 4. Number Theory 1

Lecture Note

4. Number Theory 1

Number theory

study of the integars

Def $m \mid a$ iff $\exists k \; a = k \cdot m$
(m divides a)

expect $a = 0$, which any number can divide 0 ($m \mid 0$)

Suppose a-gallon jug, b-gallon jug $a \leq b$
($a = 3$) ($b = 5$)

Thm $m \mid a$ & $m \mid b$, then $m \mid$ any linear combination of a & b

{we will get any galon that divisible by m ($0 \leq$ any $\leq b$)}

State Machine

States : pairs (x, y), where x = # gall in the "a-jug"
y = " in " "b-jug"

We are going to use gallon jug to prove

start states : $(0, 0)$

Transitions : * emptying $(x, y) \to (0, y)$
$(x, y) \to (x, 0)$

* filling $(x,y) \rightarrow (a,y)$

$(x,y) \rightarrow (x,b)$

* pouring $(x,y) \rightarrow (0, x+y)$, $x+y \leq b$

$(x,y) \rightarrow (x-(b-y), b) = (x+y-b, b)$, $x+y \geq b$

$y = b-y$

$(x,y) \rightarrow (x+y, 0)$, $x+y \leq a$

$(x,y) \rightarrow (a, y-(a-x)) = (a, x+y-a)$, $x+y \geq a$

$x = a-x$

getting 4-gallon

$a=3, b=5$ $(0,0) \rightarrow (0,5) \rightarrow (3,2) \rightarrow (0,2) \rightarrow (2,0) \rightarrow (2,5)$

$(3,4)$

Pf by induction

Assume $m|a$, $m|b$

Invariant: $P(n)$ = "If (x,y) is the state after n transition, then $m|x$, $m|y$"

Base case: $(0,0)$, $m|0 \Rightarrow P(0)$ ✓

Inductive step: $n \geq 0$, show $P(n) \to P(n+1)$

Assume $P(n)$ is True.

Suppose that (x,y) state after n transitions

$P(n) \Rightarrow m|x$ & $m|y$

After another transition, each of the jugs are filled with

$0, a, b, x, y, x+y-a, x+y-b, x+y$ gallons

↓

our current stage

$m|0, m|a, m|b, m|x, m|y, \ldots \Rightarrow m$ | any of aboves

$P(n+1)$ ✓

---

def

$\gcd(a,b)$ = the greatest common divisor of a & b

$\gcd(3,5) = 1$ $\gcd(52,44) = 4$

def

a & b are relatively prime if $\gcd(a,b)=1$

Cor $\gcd(a,b)$ | any result of linear combin of a & b

Thm Any linear combination, $L = sa + tb$, of a and b with $0 \leq L \leq b$ can be reached ($\gcd(a,b)$)

$a \leq b$

wher a = 3 gal jug & b = 5 gal jug

e.g $4 = \overbrace{(-2)}^{s} \cdot 3 + \overbrace{2}^{t} \cdot 5$

$4 = \dfrac{5 \cdot 3 - 3 \cdot 5}{3 \cdot 3 - 1 \cdot 5}$

$s' > 0$

Pf

Notice $L = sa + tb = (s + mb)a + (t - ma)b$

So $\exists s', t'$ $L = s'a + b't$ with $s' > 0$

Assume $0 < L < b$ (b.c "=" are easy)

$4 = (-2) \cdot 3 + 2 \cdot 5$

$\rightarrow (1)5 \cdot 3 - (1)3 \cdot 5$

$3 \cdot 3 - 1 \cdot 5$

$s' > 0$

Algorithm

To obtain L gallon, repeat s' times $(a = 3, b = 5)$ ← we want s' to be + so i can use in algo

- fill the "a-jug"
- Pour into "b-jug" when it become full, empty it out and continue pouring until "a-jug" empty

example of getting 3 in here 's is 3

first loop: $(0,0) \to (3,0) \to (0,3)$

second loop: $(0,3) \to (3,3) \to (1,5) \to (1,0) \to (0,1)$

third loop: $(0,1) \to (3,1) \to (0,4)$

Filled the "a-jug" s' times

Suppose that "b-jug" is emptyed u time

let r be the remainder in the "b-jug"

$0 \leq r \leq b$ $\quad 0 < L < b$

$r = s'a.u.b$

$s'a$ = total input

$u.b$ = total output

we will also use $L = s'a + b't$

$r = \underbrace{s'a + t'b}_{L} - t'b - u.b$

$r = L - (t'+u)b$

we assumed r & l on the same range $0 - b$

subtracting or adding to L change that property

$t' + u \neq 0 \Rightarrow [r > 0 \vee r > b]$

$t' + u = 0 \Rightarrow u = -t' \Rightarrow r = L$ ✓

when we change $-u \to t$

$r = s'a + t'b$

$l = s'a + t'b$

There exists a unique q (quotient) and r (reminder) such that $b = qa + r$ with $0 \leq r < a$

$r = \text{rem}(b,a)$

Euclid's algorithm

Lemma $\gcd(a,b) = \gcd(\text{rem}(b,a), a)$

example: $\gcd(105,224) = \gcd(\text{rem}(224,105),105) = \gcd(14,105)$

$224 = 2 \cdot 105 + 14$

$= \gcd(\text{rem}(105,14),14) = \gcd(7,14)$

$105 = 7 \cdot 14 + 7$

$14 = 2 \cdot 7 + 0$

$= \gcd(\text{rem}(14,7),7)$

$= \gcd(0,7) = 7$

Proof $[m|a \wedge m|b] \Rightarrow [m|b-qa = \text{rem}(b,a) \wedge m|a]$

$m|a \Rightarrow m \cdot x = a$

$m|b \Rightarrow m \cdot y = b$

$b - qa$

$(my) - q(mx)$

$m(y - qx)$

If $\text{rem}(b,a) \neq 0$ then $[m|\text{rem}(b,a) = b - qa$ and $m|a]$

$\Rightarrow [m|a \wedge m|b]$

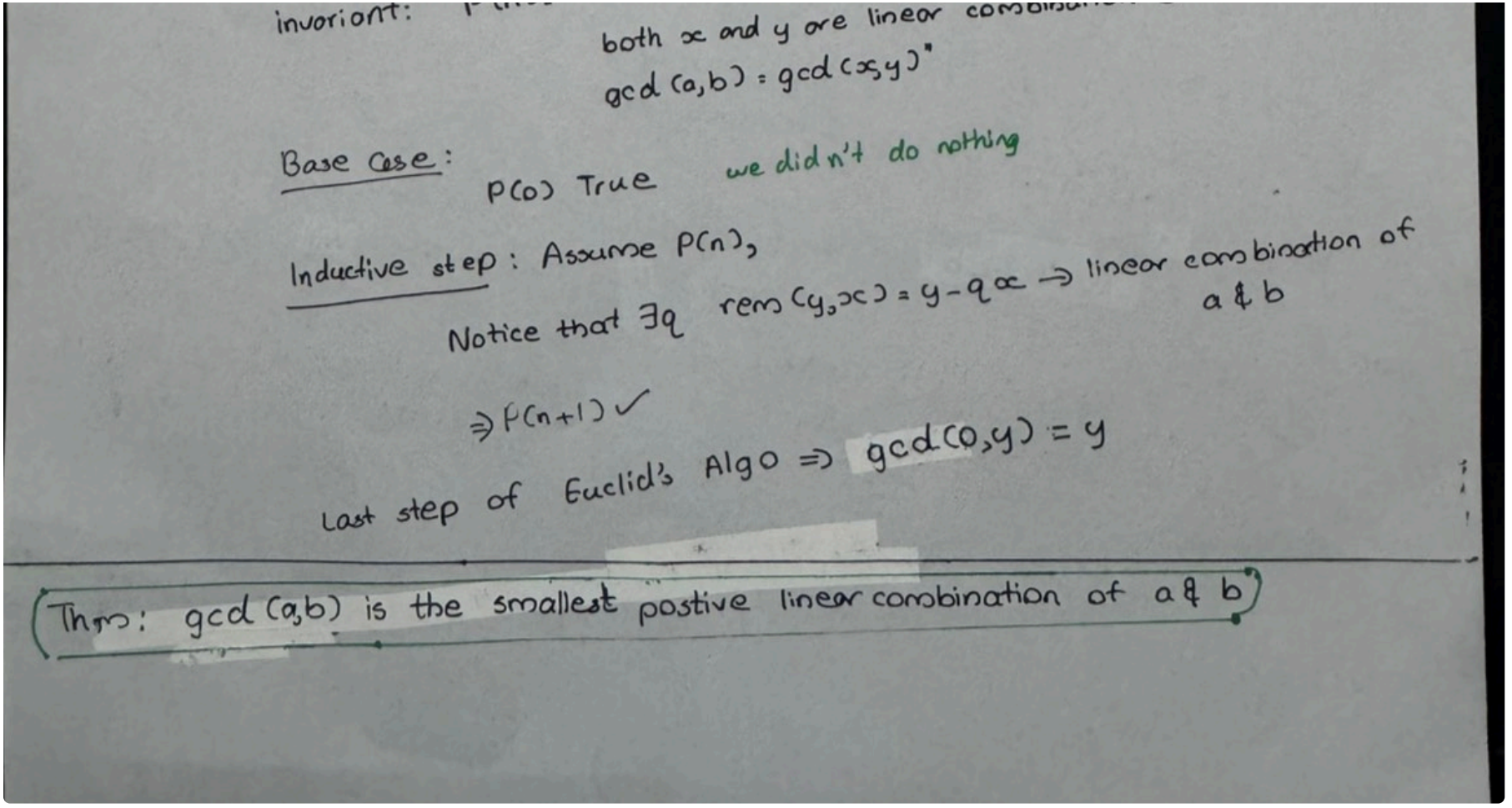
If $\text{rem}(b,a) = 0$ then $b - qa = 0 \Rightarrow b = qa$

$m|a \Rightarrow m|b$

Thm: $\gcd(a,b)$ is a linear combination of a and b

Proof: by induction

$P(n)$: "If Euclid's Algo reaches $\gcd(x,y)$ after n steps, then [illegible] of a & b.



Book Reading : 4 Number Theory

Number Theory underlies modern cryptography, which is what makes secure online communication possible.

It's also central to online commerce. Every time you buy a book from Amazon, check your grades on WebSIS, or use a PayPal account, you are relying on number theoretic algorithms.

4.1 Divisibility

a divides b iff $ak = b$ for some k

from this theory, we can drive every number divides zero. ($a \times 0 = 0$)

4.1.1 Facts about Divisibility

4.1.1 Facts about Divisibility

The lemma below states some basic facts about divisibility that are *not* difficult to prove:

Lemma 4.1.1. *The following statements about divisibility hold.*

1. *If $a \mid b$, then $a \mid bc$ for all c.*
2. *If $a \mid b$ and $b \mid c$, then $a \mid c$.*
3. *If $a \mid b$ and $a \mid c$, then $a \mid sb + tc$ for all s and t.*
4. *For all $c \neq 0$, $a \mid b$ if and only if $ca \mid cb$.*

Proof. We'll prove only part 2.; the other proofs are similar.

Proof of 2: Assume $a \mid b$ and $b \mid c$. Since $a \mid b$, there exists an integer k_1 such that $ak_1 = b$. Since $b \mid c$, there exists an integer k_2 such that $bk_2 = c$. Substituting ak_1 for b in the second equation gives $(ak_1)k_2 = c$. So $a(k_1k_2) = c$, which implies that $a \mid c$. ■

4.1.2 When Divisibility Goes Bad

Theorem 4.1.2 (division theorem):

Let n and d be integers such that $d > 0$. Then there exists a unique pair of integers q and r, such that

$$n = q.d + r \text{ and } 0 \leq r < d$$

- q : the quotient $\text{qcnt}(n, d)$
- r : the remainder of n divided by d. $\text{rem}(n, d)$

There is a remainder operator built into many programming languages.

Famous Conjectures in Number theory

Famous Conjectures in Number Theory

Fermat's Last Theorem There are no positive integers x, y, and z such that

$$x^n + y^n = z^n$$

for some integer $n > 2$. In a book he was reading around 1630, Fermat claimed to have a proof but not enough space in the margin to write it down. Wiles finally gave a proof of the theorem in 1994, after seven years of working in secrecy and isolation in his attic. His proof did not fit in any margin.

Goldbach Conjecture Every even integer greater than two is equal to the sum of two primes[2]. For example, $4 = 2 + 2$, $6 = 3 + 3$, $8 = 3 + 5$, etc. The conjecture holds for all numbers up to 10^{16}. In 1939 Schnirelman proved that every even number can be written as the sum of not more than 300,000 primes, which was a start. Today, we know that every even number is the sum of at most 6 primes.

Twin Prime Conjecture There are infinitely many primes p such that $p + 2$ is also a prime. In 1966 Chen showed that there are infinitely many primes p such that $p + 2$ is the product of at most two primes. So the conjecture is

such that $p + 2$ is the product of at most two primes. So the conjecture is known to be *almost* true!

Primality Testing There is an efficient way to determine whether a number is prime. A naive search for factors of an integer n takes a number of steps proportional to $\sqrt{n}$, which is exponential in the *size* of n in decimal or binary notation. All known procedures for prime checking blew up like this on various inputs. Finally in 2002, an amazingly simple, new method was discovered by Agrawal, Kayal, and Saxena, which showed that prime testing only required a polynomial number of steps. Their paper began with a quote from Gauss emphasizing the importance and antiquity of the problem even in his time—two centuries ago. So prime testing is definitely not in the category of infeasible problems requiring an exponentially growing number of steps in bad cases.

Factoring Given the product of two large primes $n = pq$, there is no efficient way to recover the primes p and q. The best known algorithm is the "number field sieve", which runs in time proportional to:

$$e^{1.9(\ln n)^{1/3}(\ln \ln n)^{2/3}}$$

This is infeasible when n has 300 digits or more.

4.1.3 Die Hard

problem :
On the fountain, there should be 2 jugs, do you see them? A 5-gallon and a 3-gallon. Fill one of the jugs with exactly 4 gallons of water and place it on the scale and the timer will stop. You must be precise; one ounce more or less will result in detonation. If you're still alive in 5 minutes, we'll speak.

Finding an Invariant Property

Suppose that we have water jugs with capacities a and b with $b \geq a$. The state of the system is described below with a pair of numbers (x, y), where

- x is the amount of water in the jug with capacity a
- y is the amount in the jug with capacity b.

in sample operation,

$$
\begin{aligned}
(0,0) &\to (a,0) && \text{fill first jug}\\
&\to (0,a) && \text{pour first into second}\\
&\to (a,a) && \text{fill first jug}\\
&\to (2a-b,b) && \text{pour first into second (assuming } 2a \geq b)\\
&\to (2a-b,0) && \text{empty second jug}\\
&\to (0,2a-b) && \text{pour first into second}\\
&\to (a,2a-b) && \text{fill first}\\
&\to (3a-2b,b) && \text{pour first into second (assuming } 3a \geq 2b)
\end{aligned}
$$

What leaps out is that at every step, the amount of water in each jug is of the form

$$s \cdot a + t \cdot b \tag{4.2}$$

for some integers s and t. An expression of the form (4.2) is called an *integer linear combination* of a and b, but in this chapter we'll just call it a *linear combination*, since we're only talking integers. So we're suggesting:

Lemma 4.1.3 :

Suppose that we have water jugs with capacities a and b. Then the amount of water in each jug is always a (integer) linear combination of a and b.

$a \mid b \Rightarrow b = k_1 a$

$b \mid c \Rightarrow c = k_2 b$

$= k_2(k_1 a)$

$c = a(k_2 k_1) \Rightarrow a \mid c$

Thm: from jug a & b, the amount of water in each jug is integer linear combination of a & b

Proof: by induction,

$P(n)$ ::= after n steps, the amount of water in each jug is a integer linear combination of a & b

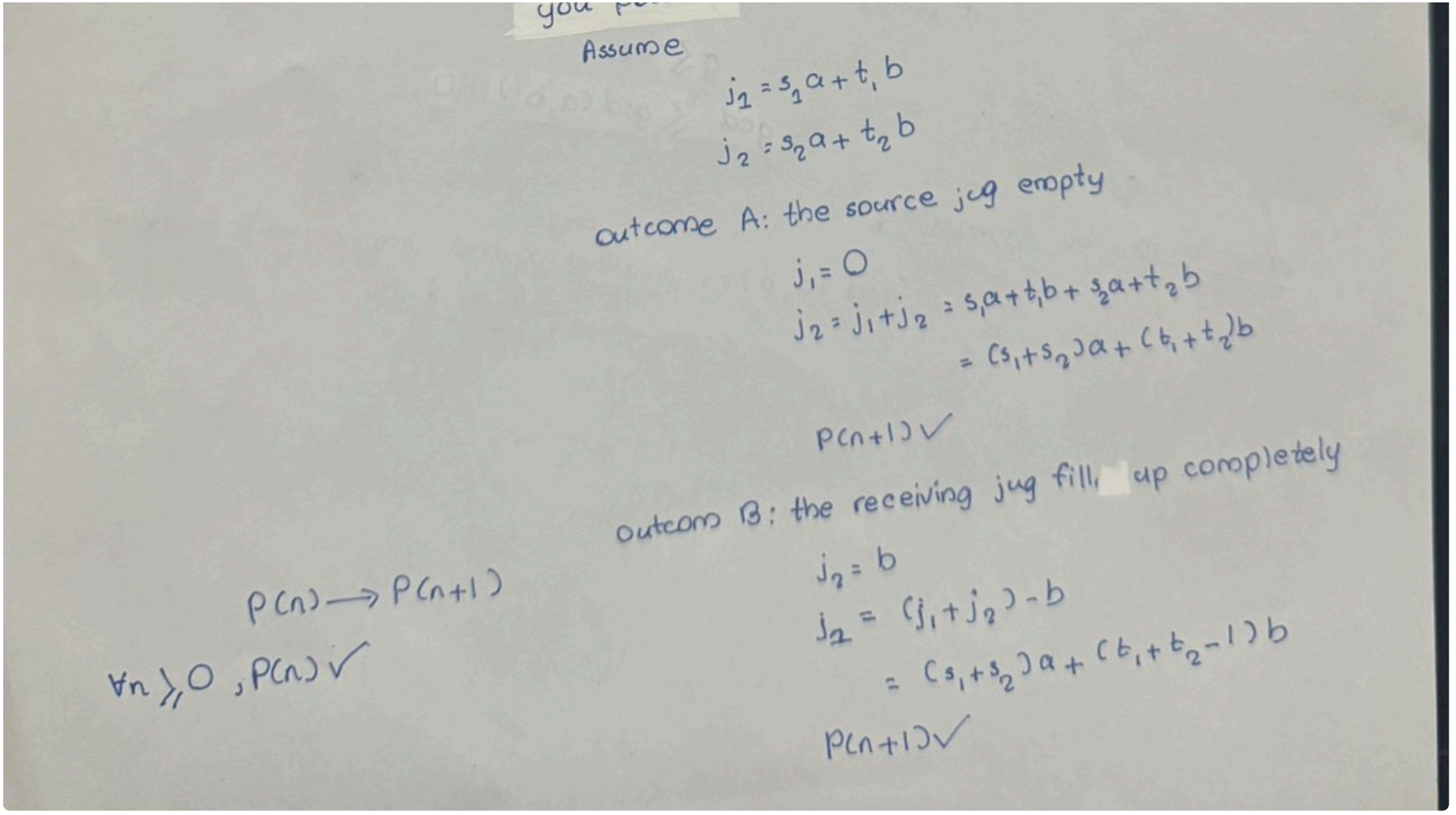
Base case: $P(0) = true$

jugs are empty at the beginning $a \cdot 0 + b \cdot 0 = 0$

Inductive step: $n \geq 0$, show $P(n) \rightarrow P(n+1)$

Assume $P(n) =$ True for the purpose of induction. There 2 possible cases at $P(n+1)$

1. filling or emptying a jug
   since it will result in $0, a, b$, which are all valid integer linear combination.
2. Pouring from one jug to another:
   pour "a" jug, j_1 to "b" jug, j_2.



Corollary 4.1.4. Bruce dies.

proof : In Die Hard 7, Bruce has water jugs with capacities 3 and 6 and must form 4 gallons of water. he needs to measure out **4 gallons**.

- Because of our lemma, *every single action* will always leave the water amounts as a linear combination of 3 and 6:

$$\text{Water} = s(3) + t(6)$$

- If you factor a 3 out of that expression, you get $3(s + 2t)$, which means **every possible water amount you can ever make must be a multiple of 3**.

- Since **4 is not a multiple of 3**, it is mathematically impossible for Bruce Willis to ever measure 4 gallons!

4.2 The Greatest Common Divisor

The greatest common divisor of a and b, $\gcd(a, b)$ is the largest number that is a divisor of both a and b.
for example $\gcd(18, 24) = 6$

4.2.1 Linear Combinations and the GCD

Theorem 4.2.1. *The greatest common divisor of a and b is equal to the smallest positive linear combination of a and b.*

For example, the greatest common divisor of 52 and 44 is 4. And, sure enough, 4 is a linear combination of 52 and 44:

$$6 \cdot 52 + (-7) \cdot 44 = 4$$

Furthermore, no linear combination of 52 and 44 is equal to a smaller positive integer.

Thm: $\gcd(a,b)$ = smallest positive integer linear combination of a & b

$$C ::= \{ sa+tb \mid sa+tb > 0 \text{ for integer } s,t \}$$

By Well Ordering Principle, there is a smallest integer, $m = s_0a + t_0b$.

Lemma 1: $\gcd(a,b) \geq m$

proof: by contrdiction, Assume for the purpose of contrdiction that "m does not evently divide a."

By Division Algo, $a = qm + r$ wher $0 < r < m$

$$r = a - qm$$

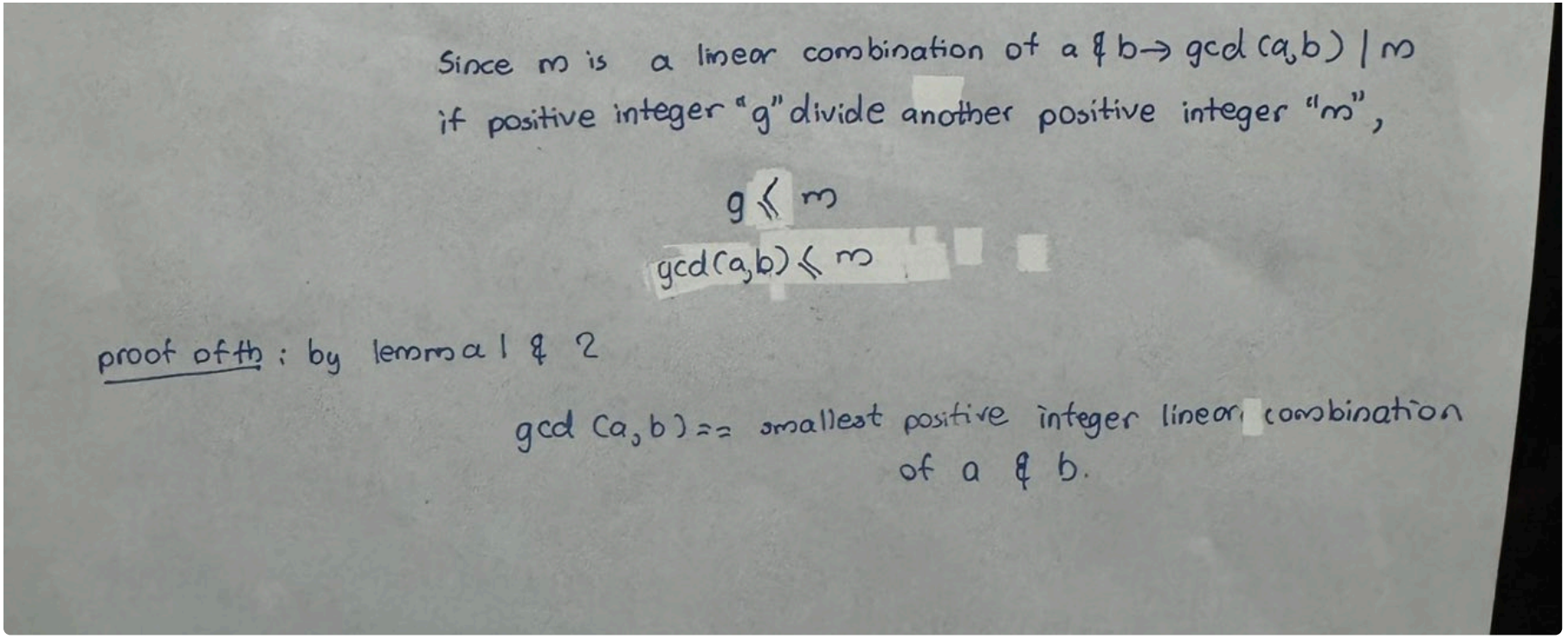
$$= a - q(s_0a + t_0b) = (1 - qs_0)a + (-qt_0)b$$

r is also linear combination which $r < m$

This is a contradiction of WOB, therfore $\gcd(a,b) \geq m$

Lemma 2: $\gcd(a,b) \leq m$

Let $g = \gcd(a,b)$, by divisibility fact $\gcd(a,b) \mid$ any integer linear combination of a & b



4.2.2 Properties of the Greatest Common Divisor

Lemma 4.2.4. *The following statements about the greatest common divisor hold:*

1. *Every common divisor of a and b divides* $\gcd(a, b)$.
2. $\gcd(ka, kb) = k \cdot \gcd(a, b)$ *for all* $k > 0$.
3. *If* $\gcd(a, b) = 1$ *and* $\gcd(a, c) = 1$, *then* $\gcd(a, bc) = 1$.
4. *If* $a \mid bc$ *and* $\gcd(a, b) = 1$, *then* $a \mid c$.
5. $\gcd(a, b) = \gcd(b, \text{rem}(a, b))$.

proof

Proof. We prove only parts 3. and 4.

Proof of 3. The assumptions together with Theorem 4.2.1 imply that there exist integers s, t, u, and v such that:

$$sa + tb = 1$$
$$ua + vc = 1$$

Multiplying these two equations gives:

$$(sa + tb)(ua + vc) = 1$$

The left side can be rewritten as $a \cdot (asu + btu + csv) + bc(tv)$. This is a linear combination of a and bc that is equal to 1, so $\gcd(a, bc) = 1$ by Theorem 4.2.1.

Proof of 4. Theorem 4.2.1 says that $\gcd(ac, bc)$ is equal to a linear combination of ac and bc. Now $a \mid ac$ trivially and $a \mid bc$ by assumption. Therefore, a divides *every* linear combination of ac and bc. In particular, a divides $\gcd(ac, bc) = c \cdot \gcd(a, b) = c \cdot 1 = c$. The first equality uses part 2. of this lemma, and the second uses the assumption that $\gcd(a, b) = 1$. ■

4.2.3 Euclid's Algorithm

$$
\begin{aligned}
\gcd(1147, 899) &= \gcd\big(899, \underbrace{\operatorname{rem}(1147, 899)}_{=248}\big) \\
&= \gcd\big(248, \underbrace{\operatorname{rem}(899, 248)}_{=155}\big) \\
&= \gcd\big(155, \underbrace{\operatorname{rem}(248, 155)}_{=93}\big) \\
&= \gcd\big(93, \underbrace{\operatorname{rem}(155, 93)}_{=62}\big) \\
&= \gcd\big(62, \underbrace{\operatorname{rem}(93, 62)}_{=31}\big) \\
&= \gcd\big(31, \underbrace{\operatorname{rem}(62, 31)}_{=0}\big) \\
&= \gcd(31, 0) \\
&= 31
\end{aligned}
$$

The calculation that gcd(1147, 899) = 31 together with Corollary 4.2.3 implies that there is no way to measure out 2 gallons of water using jugs with capacities 1147 and 899, since we can only obtain multiples of 31 gallons with these jugs.

Corollary 4.2.2. *An integer is linear combination of a and b iff it is a multiple of* $\gcd(a, b)$.

Proof. By (4.3), every linear combination of a and b is a multiple of $\gcd(a, b)$. Conversely, since $\gcd(a, b)$ is a linear combination of a and b, every multiple of $\gcd(a, b)$ is as well. ■

Now we can restate the water jugs lemma in terms of the greatest common divisor:

Corollary 4.2.3. *Suppose that we have water jugs with capacities a and b. Then the amount of water in each jug is always a multiple of* $\gcd(a, b)$.

For example, there is no way to form 4 gallons using 3- and 6-gallon jugs, because 4 is not a multiple of $\gcd(3, 6) = 3$.

4.2.4 One Solution for All Water Jug Problems

Corollary 4.2.2 says that 3 can be written as a linear combination of 21 and 26, since 3 is a multiple of $\gcd(21, 26) = 1$. In other words, there exist integers s and t such that:

$$3 = s \cdot 21 + t \cdot 26$$

We don't know what the coefficients s and t are, but we do know that they exist.

Now the coefficient s could be either positive or negative. However, we can readily transform this linear combination into an equivalent linear combination

$$3 = s' \cdot 21 + t' \cdot 26 \tag{4.4}$$

where the coefficient s' is positive. The trick is to notice that if we increase s by 26 in the original equation and decrease t by 21, then the value of the expression $s \cdot 21 + t \cdot 26$ is unchanged overall. Thus, by repeatedly increasing the value of s (by 26 at a time) and decreasing the value of t (by 21 at a time), we get a linear combination $s' \cdot 21 + t' \cdot 26 = 3$ where the coefficient s' is positive. Notice that then t' must be negative; otherwise, this expression would be much greater than 3.

Now we can form 3 gallons using jugs with capacities 21 and 26: We simply repeat the following steps s' times:

1. Fill the 21-gallon jug.

2. Pour all the water in the 21-gallon jug into the 26-gallon jug. If at any time the 26-gallon jug becomes full, empty it out, and continue pouring the 21-gallon jug into the 26-gallon jug.

At the end of this process, we must have have emptied the 26-gallon jug exactly $|t'|$ times. Here's why: we've taken $s' \cdot 21$ gallons of water from the fountain, and we've poured out some multiple of 26 gallons. If we emptied fewer than $|t'|$ times, then by (4.4), the big jug would be left with at least $3 + 26$ gallons, which is more than it can hold; if we emptied it more times, the big jug would be left containing at most $3 - 26$ gallons, which is nonsense. But once we have emptied the 26-gallon jug exactly $|t'|$ times, equation (4.4) implies that there are exactly 3 gallons left.

Remarkably, we don't even need to know the coefficients s' and t' in order to use this strategy! Instead of repeating the outer loop s' times, we could just repeat *until we obtain 3 gallons*, since that must happen eventually. Of course, we have to keep track of the amounts in the two jugs so we know when we're done. Here's the

The paragraph wraps up with a great computer science lesson: **You don't even have to calculate s' and t' ahead of time!** You don't need to sit down with pen and paper to find the coefficients. Because the math guarantees that a solution exists, you can just program a loop that says: *"Keep doing these pours until the water level hits 3 gallons, and stop"*.

4.2.5 The Pulverizer

We have shown that no matter which pair of numbers a and b we are given, there is always a pair of integer coefficients s and t such that

$$\gcd(a, b) = sa + tb$$

Unfortunately, the proof was nonconstructive: it didn't suggest a way for finding such s and t.

> kuttak", which means "The Pulverizer", Extended Euclidean Algorithm can also find integers x and y to express their greatest common divisor (gcd) as a linear combination of numbers.

The Pulverizer goes through the same steps, but requires some extra bookkeeping along the way: as we compute $\gcd(a, b)$, we keep track of how to write each of the remainders (49, 21, and 7, in the example) as a linear combination of a and b (this is worthwhile, because our objective is to write the last nonzero remainder, which is the GCD, as such a linear combination). For our example, here is this extra bookkeeping:

x	y	$(\text{rem}(x, y))$	$=$	$x - q \cdot y$
259	70	49	$=$	$259 - 3 \cdot 70$
70	49	21	$=$	$70 - 1 \cdot 49$
			$=$	$70 - 1 \cdot (259 - 3 \cdot 70)$
			$=$	$-1 \cdot 259 + 4 \cdot 70$
49	21	7	$=$	$49 - 2 \cdot 21$
			$=$	$(259 - 3 \cdot 70) - 2 \cdot (-1 \cdot 259 + 4 \cdot 70)$
			$=$	$\boxed{3 \cdot 259 - 11 \cdot 70}$
21	7	0		

The final solution is boxed.

we will treat 259 & 70 as variable x, y in $x - q \times y$ equation. so i will hold until the last

4.3 The Fundamental Theorem of Arithmetic

4.3 The Fundamental Theorem of Arithmetic

We now have almost enough tools to prove something that you probably already know.

Theorem 4.3.1 (Fundamental Theorem of Arithmetic)**.** *Every positive integer n can be written in a unique way as a product of primes:*

$$n = p_1 \cdot p_2 \cdots p_j \qquad (p_1 \leq p_2 \leq \cdots \leq p_j)$$

Notice that the theorem would be false if 1 were considered a prime; for example, 15 could be written as $3 \cdot 5$ or $1 \cdot 3 \cdot 5$ or $1^2 \cdot 3 \cdot 5$. Also, we're relying on a standard convention: the product of an empty set of numbers is defined to be 1, much as the sum of an empty set of numbers is defined to be 0. Without this convention, the theorem would be false for $n = 1$.

There is a certain wonder in the Fundamental Theorem, even if you've known it since you were in a crib. Primes show up erratically in the sequence of integers. In fact, their distribution seems almost random:

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, \ldots$$

Basic questions about this sequence have stumped humanity for centuries. And yet we know that every natural number can be built up from primes in *exactly one way*. These quirky numbers are the building blocks for the integers.

The Fundamental Theorem is not hard to prove, but we'll need a couple of preliminary facts.

Lemma 4.3.2. *If p is a prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. The greatest common divisor of a and p must be either 1 or p, since these

Proof. The greatest common divisor of a and p must be either 1 or p, since these are the only positive divisors of p. If $\gcd(a, p) = p$, then the claim holds, because a is a multiple of p. Otherwise, $\gcd(a, p) = 1$ and so $p \mid b$ by part (4) of Lemma 4.2.4. ■

A routine induction argument extends this statement to:

Lemma 4.3.3. *Let p be a prime. If $p \mid a_1 a_2 \cdots a_n$, then p divides some a_i.*

Now we're ready to prove the Fundamental Theorem of Arithmetic.

Proof. Theorem 3.1.2 showed, using the Well Ordering Principle, that every positive integer can be expressed as a product of primes. So we just have to prove this expression is unique. We will use Well Ordering to prove this too.

The proof is by contradiction: assume, contrary to the claim, that there exist positive integers that can be written as products of primes in more than one way. By the Well Ordering Principle, there is a smallest integer with this property. Call this integer n, and let

$$\begin{aligned} n &= p_1 \cdot p_2 \cdots p_j \\ &= q_1 \cdot q_2 \cdots q_k \end{aligned}$$

be two of the (possibly many) ways to write n as a product of primes. Then $p_1 \mid n$ and so $p_1 \mid q_1 q_2 \cdots q_k$. Lemma 4.3.3 implies that p_1 divides one of the primes q_i. But since q_i is a prime, it must be that $p_1 = q_i$. Deleting p_1 from the first product and q_i from the second, we find that n/p_1 is a positive integer *smaller* than n that can also be written as a product of primes in two distinct ways. But this contradicts the definition of n as the smallest such positive integer. ■